Cornell Electric Vehicles – Mechanical

Acceleration Pedal

Spring 2026 Technical Report

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General Notes:

- Don't write in first person (eg. don't say "We did this…" instead write "This was done...")
- Write in paragraph form / complete sentences but make sure to be concise
- Make sure everything makes sense to people not working on your project (eg. if a new member were to read your report for onboarding, would they understand your system / your design decisions?)

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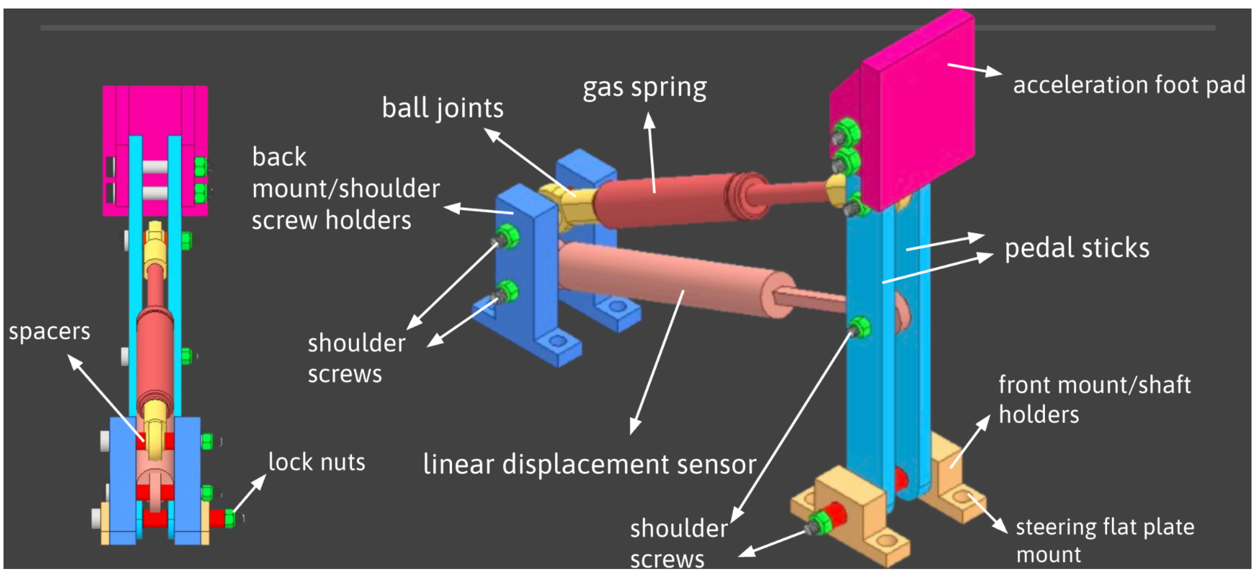
Summary

System Description

The acceleration pedal assembly is the driver input system used to control acceleration for the vehicle. Its purpose is to translate the driver's foot motion into a measurable pedal displacement while also providing enough resistance for the pedal to feel controlled and intentional. The assembly is designed to provide approximately 20–30 lb of feedback force and uses a linear displacement sensor to measure how far the pedal is pressed. This sensor data can then be used by the vehicle's electrical and controls systems to determine the driver's acceleration.

The system consists of an acceleration foot pad, two pedal sticks, front and rear mounting brackets, shoulder screws, spacers, lock nuts, ball joints, a gas spring, and a linear displacement sensor. When the driver presses the foot pad, the two pedal sticks rotate about the front pivot. The gas spring provides the return force and pedal feedback, while the linear displacement sensor tracks the motion of the pedal. In the final design, shoulder screws and flanged sleeve bearings are used at the front mount so the pedal can rotate smoothly while remaining constrained in the assembly.

Within the full car assembly, the acceleration pedal mounts to the steering flat plate and is positioned to the right of the brake pedal. The two-pedal-stick design also helps support the foot pad more evenly than a single-stick design while keeping the gas spring and sensor connected to the pedal motion path.



Terminology

- Acceleration Foot Pad – The surface that the driver presses with their foot. It is attached to the pedal sticks and moves with the rest of the pedal assembly.
- Pedal Sticks – The two vertical members that support the foot pad and rotate about the front pivot point when the pedal is pressed.
- Feedback Force – The resistance felt by the driver when pressing the pedal. The acceleration pedal is designed to provide about 20–30 lb of feedback force.

- Gas Spring – The component that provides the pedal’s resistance and return force.
- Linear Displacement Sensor – The sensor used to measure how far the acceleration pedal has been pressed.

Research and Requirements

Background Research

The acceleration pedal is used to control how much the car accelerates. Since the pedal does not mechanically connect to the motor, it needs a sensor to detect how far the driver is pressing it. It also needs a spring or similar component to give the driver feedback force, so the pedal does not feel loose. For this design, a linear displacement sensor was used to collect pedal position data, and a gas spring was used to provide feedback force and return the pedal. The design also had to sit to the right of the brake pedal and mount to the steering flat plate.

Design Requirements

The acceleration pedal needs to give the driver a controlled way to accelerate the car. It should provide around 20–30 lb of feedback force and use a displacement sensor to measure how far the pedal is pressed. The pedal also needs to mount to the steering flat plate and be positioned to the right of the brake pedal. For this semester, the design was changed from a one-pedal-stick design to a two-pedal-stick design.

Design Requirements Derived from Rules

There is no direct Shell Eco-marathon rule that requires an acceleration pedal. However, the pedal still needs to be designed safely because it is part of the driver controls. It should not interfere with the brake pedal, steering system, or driver position.

Key Parameters

The most important parameter for the acceleration pedal is the feedback force. The target feedback force is 20–30 lb. Another important parameter is the pedal motion, which is limited by the gas spring so that the displacement sensor does not take the load. The pedal also has to fit on the steering flat plate and stay to the right of the brake pedal.

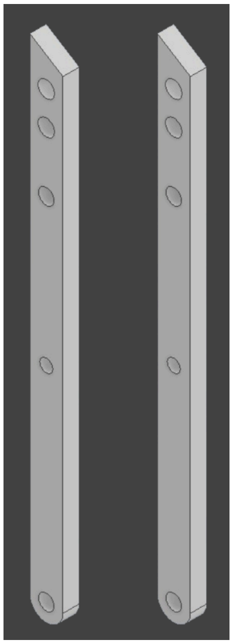
Selected Design and Justification

The selected design uses a linear gas spring, a linear displacement sensor, and a two-pedal-stick layout. This design was chosen because it is simple, cost-effective, and gives control over both pedal feedback force and pedal position data. The gas spring gives the driver feedback force, while the displacement sensor measures how far the pedal is pressed. A gas spring was chosen instead of a torsion spring because it is easier to mount and can help limit pedal travel. A torsion spring could work, but it would need a separate hard stop and would be harder to design. The linear displacement sensor was also reused from last year, which saved cost. The two-pedal-stick design was chosen because it distributes the driver's input force across both pedal sticks, supports the pedal pad more evenly, and makes the pedal feel sturdier with less wobble than a one-stick design.

Detailed Design (CAD)

Design Overview

Acceleration Pedal Stick

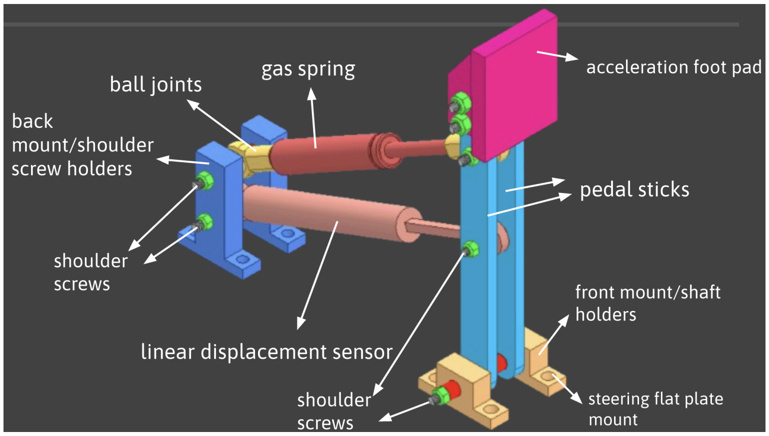


The acceleration pedal stick was one of the most important parts of the design because it controls the pedal motion, feedback force, and sensor travel. My final design uses a two-pedal-stick layout instead of the one-pedal-stick design used previously. The acceleration foot pad is mounted at the top of the two pedal sticks, while the bottom hole acts as the main pivot point for the pedal. The other holes are used to connect the gas spring and the linear displacement sensor. This layout allows the gas spring to provide feedback force while the displacement sensor tracks pedal travel without being used as the hard stop for the system.

The hole placement was chosen so that the gas spring and sensor move through the correct range as the pedal rotates. The pedal stick design showed separate connection points for the acceleration foot pad, gas spring, linear displacement sensor, and main pivot. Analysis also showed that the gas spring should limit the pedal motion instead of the sensor, so the sensor does not bottom out when the pedal is fully pressed or released. The two-stick design also supports the foot pad more evenly and makes the assembly more stable because the load from the driver's foot is shared between both pedal sticks.

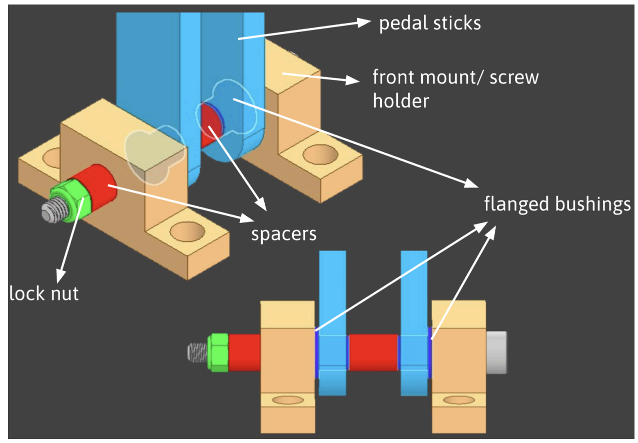
Acceleration Pedal Assembly

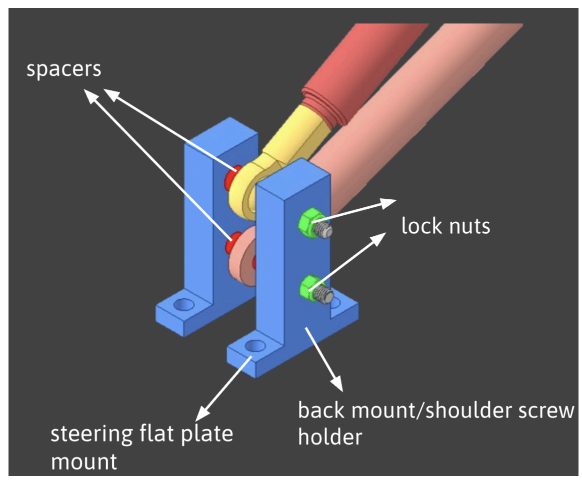
The full acceleration pedal assembly includes the acceleration foot pad, two pedal sticks, front mount, back mount, gas spring, linear displacement sensor, ball joints, shoulder screws, bushings, spacers, and lock nuts. The main purpose of the assembly is to give the driver a smooth pedal input while also sending pedal displacement data. The design requirements were that the pedal should provide about 20–30 lb of feedback force, include a displacement sensor, interface with the steering flat plate, and be positioned to the right of the brake pedal.



The gas spring is mounted between the pedal sticks and the back mount using ball joints. As the driver presses the acceleration pedal, the pedal sticks rotate about the lower pivot and compress the gas spring. This creates the feedback force that the driver feels. The linear displacement sensor is mounted separately so it can measure how much the pedal is pressed. A key design decision was making sure that the gas spring limits the range of motion instead of the sensor. This protects the sensor from being overloaded or during pedal travel.

Acceleration Shaft Assembly





The acceleration shaft assembly is the pivot system that allows the pedal sticks to rotate smoothly when the driver presses the pedal. In the CDR design, the pedal rotated about a shaft located in the front mount. At that stage, the design assumed that no bearing was needed because the pedal would only rotate a few degrees back and forth. However, this was updated in the FDR to improve the reliability of the pivot.

In the final design, the shafts were replaced with shoulder screws, and flanged sleeve bearings/bushings were added to the front mount. The shoulder screws provide a more controlled and consistent pivot surface than a custom shaft, while the bushings reduce friction and wear between the pedal sticks and the mount as they are made from aluminum. Bushings were chosen instead of full bearings because the pedal only rotates through a limited range of motion, so a rolling bearing was not necessary.

The shaft assembly also uses spacers and lock nuts to keep the two pedal sticks aligned and properly constrained. This prevents side-to-side motion while still allowing the pedal to rotate freely. The FDR design also uses shoulder screws at the back mount for the gas spring and displacement sensor connections.

Off-the-shelf Component Selection

For this acceleration pedal design, I used several off-the-shelf components because they were either too difficult to manufacture or not worth the time to make. The main off-the-shelf parts were the displacement sensor, gas springs, ball joints, shoulder screws, lock nuts, bushings, and aluminum stock. The total BOM cost was about $189.24, but the displacement sensor was reused from last year.

The **displacement sensor** was reused from last year's acceleration pedal. This part is needed to measure how far the pedal is being pressed.

The **gas springs** were bought from McMaster for $18.30 each, or $36.60 total. They are needed to give the pedal feedback force. They also help limit the pedal motion so the displacement sensor does not bottom out.

The **ball joints** were bought from McMaster for $8.88 each, or $17.76 total. They connect the gas spring and displacement sensor to the pedal assembly and allow the parts to rotate slightly as the pedal moves.

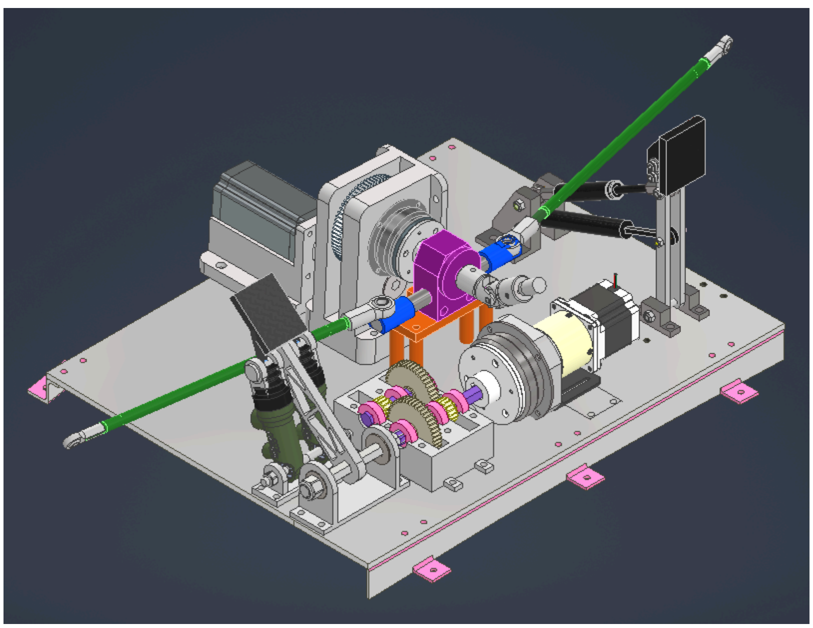
The **shoulder screws** were bought instead of machining custom shafts. This was one of the main changes from my CDR to FDR design. They make the assembly easier because they already have a smooth surface for rotation and a threaded end for lock nuts. This saved machining time and made the design easier to assemble.

The **flanged sleeve bearing**, or bushing, was bought for $19.10 and added to the front pivot. I originally did not include a bushing in the CDR because the pedal only rotates a small amount, but I added it in the FDR to make the pedal rotate more smoothly and reduce wear.

The **lock nuts** and **6061 aluminum stock** were also bought because they are standard parts. The lock nuts keep the system from loosening, and the aluminum stock was used for the machined mounts because it is strong, lightweight, and easy to machine.

Interfacing

The acceleration pedal assembly will mount directly to the steering flat plate using 1/4-20 screws. The front mount and back mount both have base holes that line up with the steering flat plate, so the full pedal assembly can be bolted down securely. The acceleration pedal is shown on the right side in the image below.

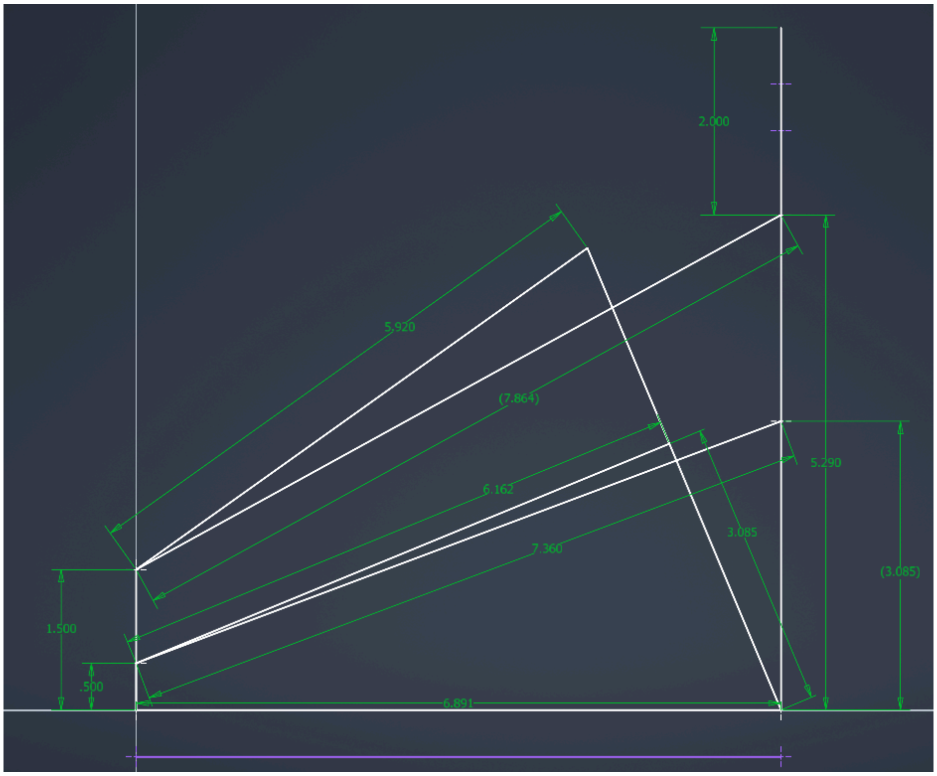


Analysis

The acceleration pedal must withstand about 20–30 lb of driver input force, with heat and high voltage not being major concerns.

Hand Calculations

The main hand calculation I did was a static force analysis of the acceleration pedal. The linear displacement sensor was not included in the force calculation because it should not take the main load. Instead, the gas spring provides the feedback force and limits the pedal travel. Using the pedal stick hole locations, gas spring force, and moment balance about the main pivot, I could estimate the feedback force felt at the pedal pad.



A similar calculation was also done in MATLAB using the pedal geometry. One 20 N gas spring gave a maximum feedback force of about 19.69 lb and a maximum pedal angle of 61.16 degrees. This showed that one gas spring was close to the desired 20 lb feedback force, so the final design used one gas spring instead of two.

The pedal geometry was also checked in CAD using the dimensions of reused displacement sensor, gas spring, and pedal stick hole locations. This helped make sure the gas spring and sensor could move through the pedal's range of motion without interfering with each other.

untitled.m

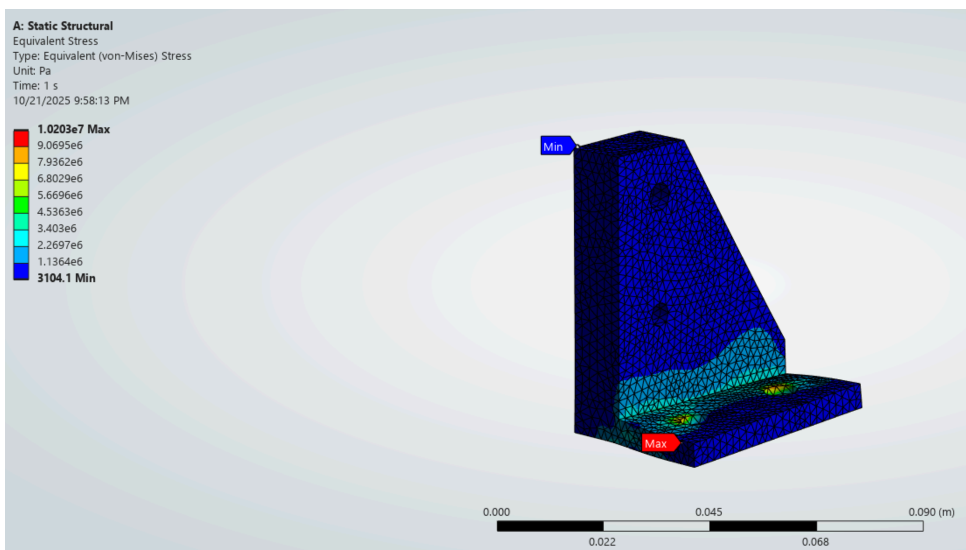
/MATLAB Drive/untitled.m

```
% acc. pedal
% gas spring
l=5.313;
L=6.44;
%this is supposed to be my mm to inch convertion...
hatThickness= 1.1811;
springExtended=5.51+hatThickness*2;
springStroke=1.97;
springCompressed=3.54+hatThickness*2;
springForce=27;
d=sqrt(springExtended*springExtended-l*l);
minThetaSpring=acosd((d*d+l*l-springCompressed*springCompressed)/(2*l*d));
feedbackForceMin=((springForce*l*l)/springExtended)/L;
del=acosd((l*l+springCompressed*springCompressed-d*d)/(2*l*springCompressed));
feedbackForceMax=(springForce*l*cosd(90-del))/L
k=1;
```

ANSYS

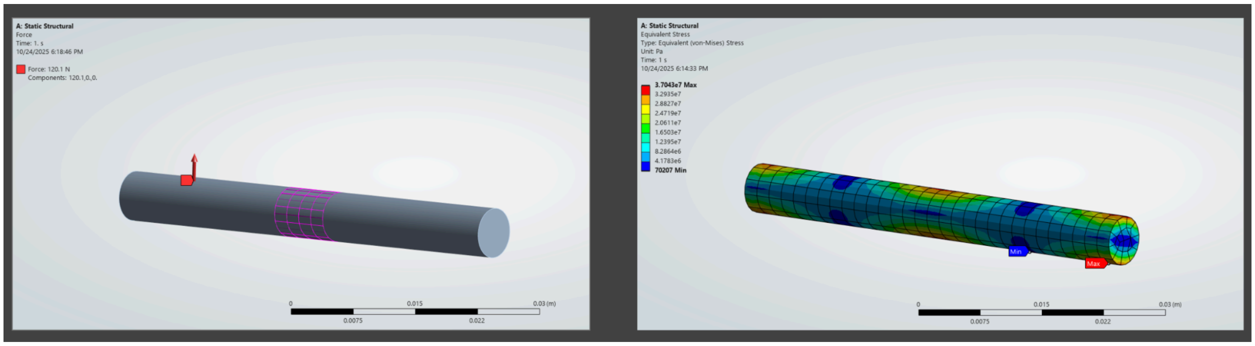
Boundary Conditions

For the ANSYS analysis, I modeled the main load cases on the back mount and the M6 gas spring shaft from my CDR design. The back mount was fixed at the bottom mounting holes, which represents where it bolts into the steering flat plate. A force was applied at the hole where the gas spring connects to the mount. This simulates the load from the gas spring pushing on the mount during pedal use.



For the M6 gas spring shaft analysis, the shaft was supported at the contact points where it would sit in the mount, and the gas spring force was applied near the center where the ball joint connects. This was done to check if the shaft could handle the bending load from the gas spring. The analysis gave a maximum stress of 37.04 MPa, which led to a factor of safety of about 7.42.

These boundary conditions are a simplified version of the real system because they do not fully model every spacer, lock nut, or surface. Also, the final FDR design changed from custom shafts to shoulder screws, so this analysis is more of a conservative check.



Mesh and Convergence

This wasn't something you considered in my analysis.

Results and Optimization

I used ANSYS to check where the highest stresses would occur in the pedal stick, back mount, and gas spring shaft. The main design changes after analysis were simplifying the mount shape for easier machining, replacing the custom shafts with shoulder screws, and adding bushings at the front pivot. The final safety factors from my analysis were high enough for the expected pedal loading. The back mount had a factor of safety of about 27.1, the M6 gas spring shaft had a factor of safety of about 7.42, and the pedal stick had a factor of safety of about 37.93.

Bill of Materials

Name	Retailer	Quantity	Unit Price ($)	Price ($)
Displacement Sensor	AliExpress	1	31.80	31.80
Gas Spring	McMaster	2	18.30	36.60
Ball Joints	McMaster	2	8.88	17.76
Pedal Stick	Send cut Send	2	9.82	19.64
Acceleration Pad Screws	McMaster	2	2.14	4.28
Acceleration Pad Lock nut	McMaster	2	4.36 for 100 pack	4.36
M4 x 0.7 mm Lock	McMaster	2	$5.03 per pack of	5.03

nuts			100	
M5 x 0.8mm Lock nuts	McMaster	3	$5.52 per pack of 100	5.52
Pedal Stick potentiometer shoulder screw	McMaster	1	3.63	**3.63**
Pedal Stick gas spring screw	McMaster	1	1.87	1.87
Back Mount Potentiometer Screw	McMaster	1	17.36	17.36
Back Mount Gas Spring Screw	McMaster	1	4.07	4.07
Flanged Sleeve Bearing	McMaster	1	19.1	19.1
Point of Rotation Screw	McMaster	1	4.35	4.35
Aluminum 6061 Stock	McMaster	1	13.87	13.87
			Total:	189.24

Manufacturing Plan

Part(s)	**Manufacturing Processes**
Front mount	Manual mill
Back mount	Manual mill
Shoulder Screw	Off-the-shelf
Bushings	Off-the-shelf
Spacers	3D-Print
Lock Nuts	Off-the-shelf
Gas Spring	Off-the-shelf

Ball Joint	Off-the-shelf
Acceleration Foot Pad	3D-Print
Displacement sensor	Reused from last year

The front and back mounts were manually milled from aluminum stock. The main operations were cutting the stock to size, facing the part, drilling the mounting holes, and drilling the holes for the shoulder screws. The most important tolerances were the hole locations for the pivot, gas spring, and displacement sensor because these control the pedal motion and alignment. The two pedal sticks were made through SendCutSend because they are flat plate parts with several holes. Everything else was off-the-shelf, which reduced machining time.

Testing Plan

Testing for the acceleration pedal should be done once the full assembly is mounted to the steering flat plate. The main test is to press the acceleration pedal by hand and check that the pedal rotates smoothly, returns to its starting position. The feedback force should also be tested to make sure the pedal feel is close to the target range of 20–30 lb. The displacement sensor also needs to be tested to make sure it measures pedal travel correctly and does not bottom out at either end of motion. This is important because the gas spring should limit the pedal travel, not the sensor.

Project Reflection and Future Work

Overall, the acceleration pedal design improved a lot from the CDR to the FDR. The final version used shoulder screws instead of custom shafts, added bushings at the pivot, and used simpler front and back mounts that were easier to machine. One important lesson from this project was to pay attention to any areas where aluminum could rub against aluminum. The acceleration pedal needs to rotate smoothly, but it also needs to feel sturdy and not have wobble, so bushings and proper spacing are important.

If I were to start the project over, I would focus on manufacturability much earlier. At first, I designed more around the ideal geometry, but later I had to change parts so they were easier to machine and assemble. The pedal stick could also be improved in a future iteration. The current design works, but the geometry could be cleaned up to make it lighter and more aesthetically pleasing.

For testing, I would do more testing before finalizing the design. The pedal should be tested for feedback force, smooth motion, and wobble. It would also be useful to test different gas springs to see if another option gives a better pedal feel while still keeping the correct range of motion.

References

https://www.aliexpress.us/item/3256804815094978.html?spm=a2g0s.imconversation.0.0.7bdf3e5fPdJZz1&_gl=1*107qt19*_gcl_au*MTY3NjY1NzE2NS4xNzI2MDAyMzE1*_ga*NzgwNjU0NjUwMTI2MTc2LjE3MjYwMDIzMTA1NDk.*_ga_VED1YSGNC7*MTcyNjMyMzYxMy4zLjEuMTcyNjMyMzg5MS42MC4wLjA.&gatewayAdapt=glo2usa4itemAdapt

https://docs.google.com/spreadsheets/d/1ETC8fwLGo2M1XecAJ__50IpQtGJ5c0CoHm-GSNiau84/edit?usp=sharing